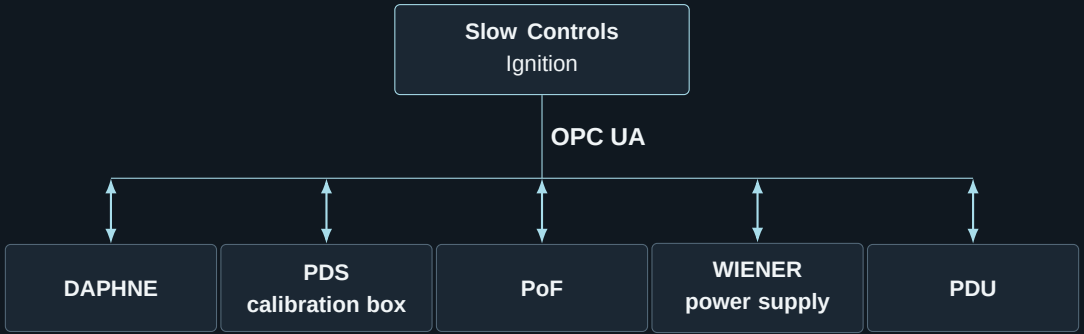
SC–PDS–DPS interface

Control, monitoring and protection

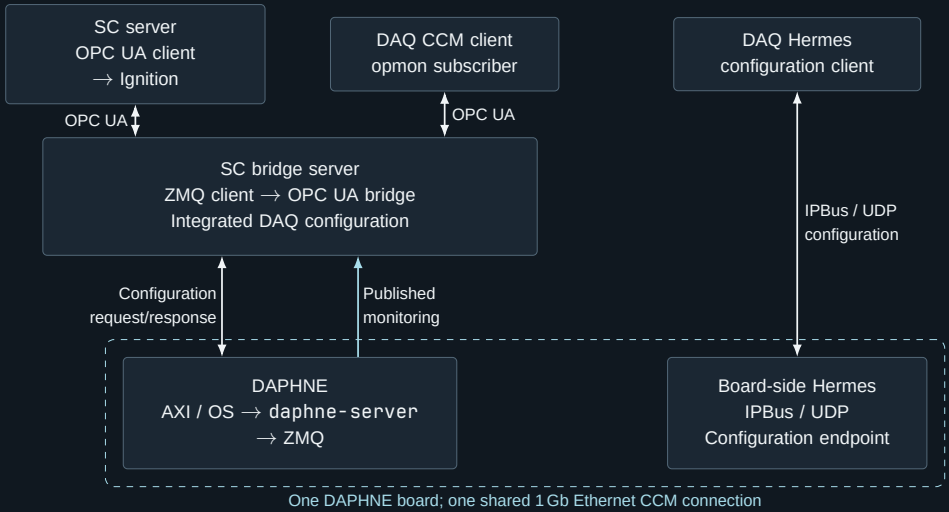
PDS	Equipment, native services and operating limits.
SC	OPC UA integration, permissions, health and availability.
DPS	Agreed protection, independent of SC and DAQ.
DAQ	Readout configuration, acquisition and use of SC availability.

Every system reaches SC through OPC UA



Commands use request/response. Monitoring is published.

DAPHNE control path

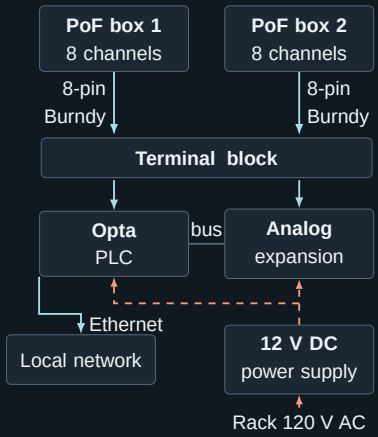


Keep the variables we can read

System	Control	Publish
DAPHNE	Applied DAQ settings	Hardware V/I/P/T, bias and AFE state
PoF	—	Team plan: 4–7 V + 1 chassis RTD
PDS calibration box	Pulse settings, bias and enable	Status and monitor readbacks
WIENER PS	Voltage demand and output switch	Measured V/I, status; power reader pending
PDU	Outlet switch	Power, outlet state and fitted sensors

OPMON publishes data counters and Hermes queries.

PoF: the team’s proposed topology



Two boxes share one controller

One Opta + one analog expansion.

Monitoring per box

4–7 laser voltages.
One chassis temperature (RTD).

SC through OPC UA

Reported updates: about 1 Hz.

Rack draft: 40 mini-racks,
80 boxes, 640 laser drivers.

Rack monitoring: 44 PDUs, 440 outlets

Draft allocation: one PDU per rack.

Rack group	PDU model	PDUs	Outlets
40 top-plate mini-racks	PX3-5147R	40	320
4 PDS power racks	PX3-5170V-E2F5	4	120
Total		**44**	**440**

Draft profile per rack: 4 temperatures, 1 leak detector, 1 smoke detector.

176 temperature probes, 44 leak detectors, 44 smoke detectors.

Define each SC point before using it

1. **Source:** device, channel, NodeId, units and scaling.
2. **Access:** Control or Publish, with an owner.
3. **Limits:** alarm levels, hysteresis and on-delay.
4. **Handling:** scan/history rate, stale data and response.

Unknown values stay blank. Protection actions need agreement with DPS.

The draft contains 159 variable rows

Tab	Total	Control	Publish	Pending
DAQ	53	49	4	0
OPMON	28	0	28	0
SC	27	0	27	0
PoF	16	0	16	16
PDS calibration box	19	7	12	0
WIENER	6	2	4	6
PDU	10	1	9	0
Total	**159**	**59**	**100**	**22**

Next: map PoF and WIENER, verify PDU assignments, and export existing readers.

FD-VD draft subtotal: 20,804 variables

42 boards × 32 active channels; 84 data links.

PoF count uses the 8 T + 8 V + one enable working case.

Interface	Control	Publish	Subtotal
DAQ	9,618	168	9,786
OPMON	0	4,914	4,914
SC	0	1,554	1,554
PoF	80	1,280	1,360
PDS calibration box	584	512	1,096
WIENER	116	174	290
PDU	440	1,364	1,804
Total allocated	**10,838**	**9,966**	**20,804**

Add 6 per Hermes input buffer, 12 per populated HD-mezzanine AFE,

and 1 per defined WIENER power-consumption point.

Backup: Beckhoff I/O per PoF box

Reuse the Beckhoff PLC and flange-side EtherCAT connection.

Signal	Points	Example terminal	Modules
Temperature inputs	8	EL3208: 8 RTD inputs	1
Voltage inputs	8	EL3068: 8 analog inputs	1
Box enable/disable	1 output	EL1859: use 1 of 8 outputs	1
Eight-channel draft case			**3**

One enable controls the whole box. Readout: 2 terminals; box control adds 1.

Backup: Beckhoff sizing for 40 mini-racks

Working I/O case: 8 T + 8 V + one enable per box.

Quantity	Per mini-rack	40 mini-racks
PoF boxes	2	80
Laser drivers	16	640
Temperature inputs	16	640
Voltage inputs	16	640
Box-enable outputs	2	80
Signal points	**34**	**1,360**
I/O terminals	**5**	**200**

Per rack: 2 EL3208 + 2 EL3068 + 1 shared EL1859.

Backup: design and implementation differ

	ICD draft 0.2 design	Selected implementation
DAPHNE commands	DAQ and SC use bridge arbitration.	DAQ has direct ZeroMQ calls; V3 FE and clock commands are separate.
Monitoring	Board publishes; bridge supplies SC and opmon.	Existing reads/queries do not establish this publication path. Some readers are local only.
OPMON scope	Data monitoring reaches DAQ through the bridge.	This inventory removes hardware V/T from OPMON; code migration remains.
Other systems	Calibration and PoF use SC adapters.	Calibration register client and PX3 bindings exist. PoF/WIENER mapping and SC exports remain open.

Backup: sources

- SC–PDS ICD: local draft 0.2, §§1–3; revision `bfb2a8b`. DAPHNE figure copied verbatim.
- `daphne-os develop edc61f1`; `daphnemodules develop ba6fd3c`; local Hermes `d7de1de`.
- PDS calibration-box register client and register map v4.2a.
- WIENER guide; Raritan submittals; PDU definitions and sensor profile from DocDB 32655 v8.
- PoF team presentation, 14 October 2026; rack draft, SC plan p. 4 and Beckhoff datasheets.
- SC intake template; operational workbook v1.2 with system inputs and indexed-point expansion.

Draft for discussion. No live hardware or protection qualification.